

Total No. of Questions : 5]

SEAT No. :

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F.Y. M.C.A. (Management)

RMW 554 MJ - RM21 : RESEARCH METHODOLOGY

(2024 Pattern) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) *Draw neat diagram wherever necessary.*
- 2) *All Questions carries equal marks.*
- 3) *Scientific calculator is allowed.*

- Q1)** a) Describe research process with suitable diagram in detail. [5]
b) Explain importance of Literature review in research. [5]

OR

- c) Differentiate qualitative and quantitative research methods with suitable example. [5]
d) What is Research Methodology? Explain objectives of research. [5]

- Q2)** a) Elaborate types of research with example. [10]

OR

- b) What is hypothesis? [5]
Prepare a hypothesis (null & alternate) for following situations.
i) Researcher wants to understand the impact of social media usage on children's behaviour.
ii) Researcher wants to understand the lectures attended by first-year students and its effect on final exam scores.
c) What are the key components of a well-structured research design? [5]

- Q3)** a) Explain Probability and Non-probability Sampling methods with suitable examples. [10]

OR

- b) Suppose you are asked to select a sample for is conducting a survey on students' learning habits. Which sampling method would be most suitable and why? What will be sample size for the same? [10]

P.T.O.

Q4) a) Discuss Chi - square test. **[10]**

From the following data, find out whether there is any relationship between Gender(Male/Female) & Colour Preferences:

Colour	Males	Females	Total
Pink	10	40	50
Black	70	30	100
Yellow	30	20	50
Total	110	90	200

For the above example Formulate null and alternate hypothesis and apply Chi-square test.

(consider : chi - squared value 0.05 significance level at degree of freedom 2 is = 5.99)

OR

b) What is t-test, When it is used? **[10]**

Determine whether the average weight of sample of 20 mangoes is significantly different from population average weight of 70 gms. The sample mean weight is 70.55gms & sample standard deviation is 2.82 gm.

Apply one sample t test for the above example (consider t value at 0.05 significance =2.093).

Q5) a) Select a research topic within a domain of your choice (e.g., Artificial Intelligence, Machine Learning, Cloud Computing, Internet of Things (IOT), Block Chain, Social Media, Cybersecurity, etc. or any topic).

Prepare a research outline which includes a Title, abstract, introduction, literature review, Objectives, Hypothesis, methodology, results & discussion, conclusion, and references. **[10]**

OR

b) Prepare an outline for a research on topic “Impact of Social Media on Indian Youth” with proper title, objectives, hypothesis, Research Gap, Data Collection, sampling, Result and Discussion and Conclusion. **[10]**

